Some basic Concept of Chemistry :-

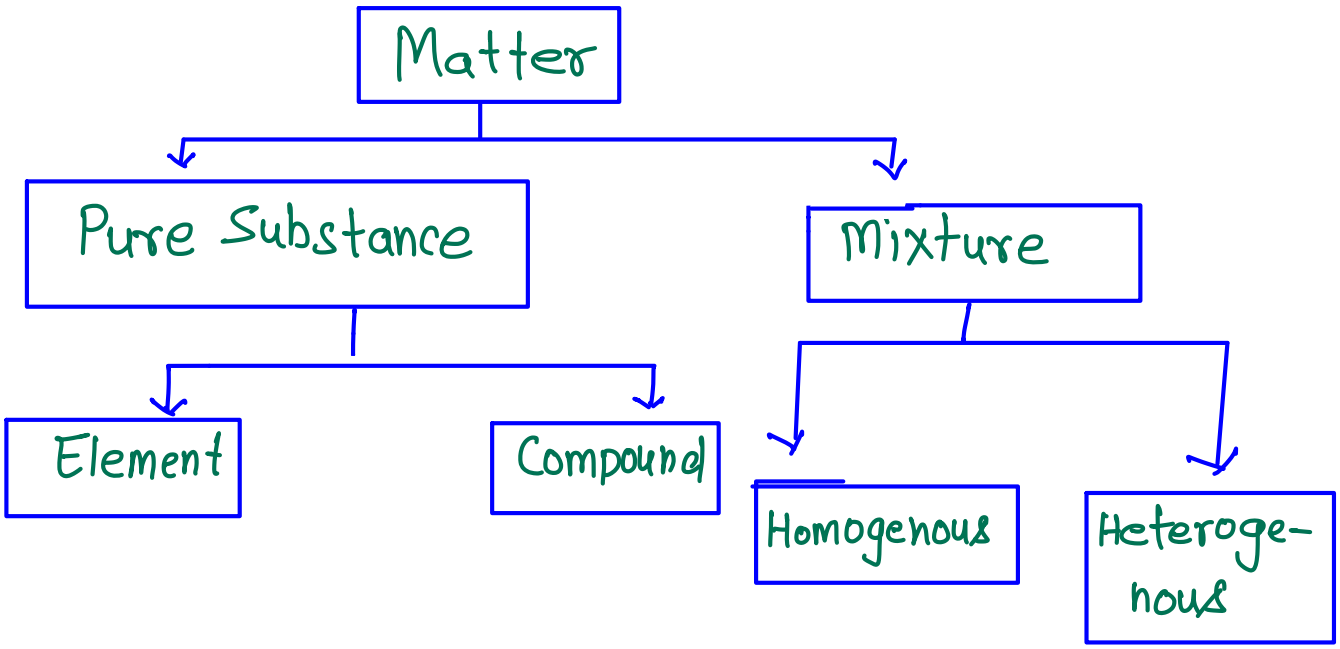
Physical Property :-

The Property which Can be measured without changing the chemical Composition of the Substance is known as physical Property like mass, volume, density etc.

Chemical Property :-

The property which Can be evaluated at the Cost of matter itself is known as chemical Property.

Classification of Matter :-



Law of chemical Combination:-

Law of Conservation of Mass:- for any chemical change total mass of active reactants are always equal to the mass of the Product formed. It is a derivation of Dalton's atomic theory 'Atoms neither Created nor destroyed'.

Total masses of reactants = Total masses of Products + Masses of unreacted reactants.

Example:- 4.90 g of $KClO_3$ when heated Produced 1.92 g of oxygen and the residure (KCl) Left behind weights 2.96 g. Show that these results illustrate the law of Conservation of mass.

Sol:- Weight of $KClO_3$ taken = 4.90 g

total weight of the Products ($KCl + O_2$) = 2.96 + 1.92 = 4.88 g

Difference between the wight of the reactant and the total weight of the Product = 4.90 - 4.88 = 0.02 g

This small difference may be due to experimental error.

Thus, Law of Conservation of mass holds good within experimental error.

Law of Constant Proportion :-

A chemical Compound always Contain Same elements in definite proportion by mass and it does not depend on the Source of Compound.

Example :- $C + O_2 \longrightarrow CO_2$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

The Composition of CO_2 obtained by different means always having Same C:O Ratio =

$$\frac{12}{32} = 0.375$$

Law of Multiple Proportion :-

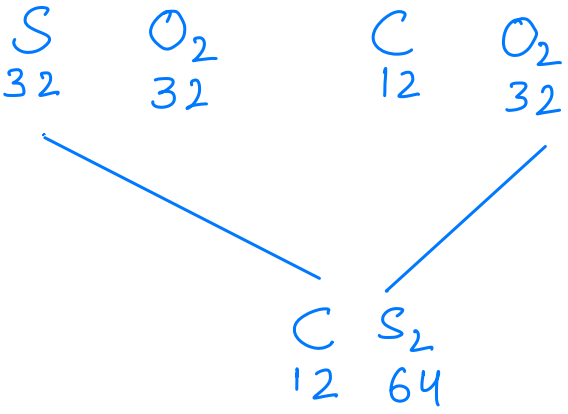
When two elements Combine to Form two or more than two different Compounds then the different masses of one element which Combine with fixed mass of the other element bear a Simple Ratio to one another. Ex :- CO, CO_2

Law of Reciprocal proportions - When two element combine separately with a fixed mass of a third element they do so in the same weight

ratio as they would combine with each other.

Example:- Ratio of masses of Carbon and Sulphur which Combine with fixed mass (32 Part) of oxygen is 12:32 or 3:8 ..... (1)

In CS_2 ratio of masses of Carbon and Sulphur is 12:64 or 3:16 ..... (2)



The two ratio (1) and (2) are related to each other by $\frac{3}{8} : \frac{3}{16}$ or 2:1 or $\frac{3}{8} : 2\left(\frac{3}{16}\right)$

First ratio is integral multiple of second.

Gay Lussac's Law of Combining Volumes:-

At given temperature and pressure the Volume of all gaseous reaction and Products bear a simple whole number ratio to each other.

Example:-

$$H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)} \quad (1:1:2)$$

One Volume of hydrogen reacts with one volume of chlorine to form two volume of HCl gas.

the ratio by Volume which gases bear is 1:1:2 which is a simple whole number ratio.

Other Example are :-

$2SO_2 + O_2 = 2SO_3$ [2:1:2]

2 Volume 1 Volume 2 Volume

$2NH_3 \rightleftharpoons N_2 + 3H_2$ [2:1:3]

2 Volume 1 Volume 3 Volume

$NH_3 + HCl \rightleftharpoons NH_4Cl$ [Vapour] [1:1:1]

1 volume 1 Volume 1 Volume

Dalton's Atomic Theory :-

In 1808, Dalton Published 'A new System of chemical philosophy' in which he proposed the following :-

→ Matter Consist of indivisible atoms.

→ All the atoms of a given element have identical Properties including identical mass. Atoms of different elements differ in mass.

→ Compound are Formed when atom of different element Combined in a fixed Ratio.

Chemical reaction involve reorganization of atom. These are neither Created or destroyed in a Chemical reaction.

Limitations:- The main failures of Dalton's atomic theory are:-

* Atom was no more indivisible. It is made up of various Sub-atomic particles like electron Proton and neutron etc.

* It failed to explain the nature of forces that bind together different atom in molecules.

* It failed to explain Gay-Lussac's law of Combining Volumes.

* It did not make any distinction between ultimate Particle of an element that takes Part in reaction (atoms) and the ultimate Particle that has independent existence (Molecule).

* Isobars Can not be explained by Dalton's theory.

Atomic mass and Molecular mass:-

Atomic mass:- The atomic mass of an element Can be defined as the number which indicates how many times the mass of....

One atom of the element is heavier in Comparision to the mass of one atom of hydrogen.

Atomic mass of an element :-

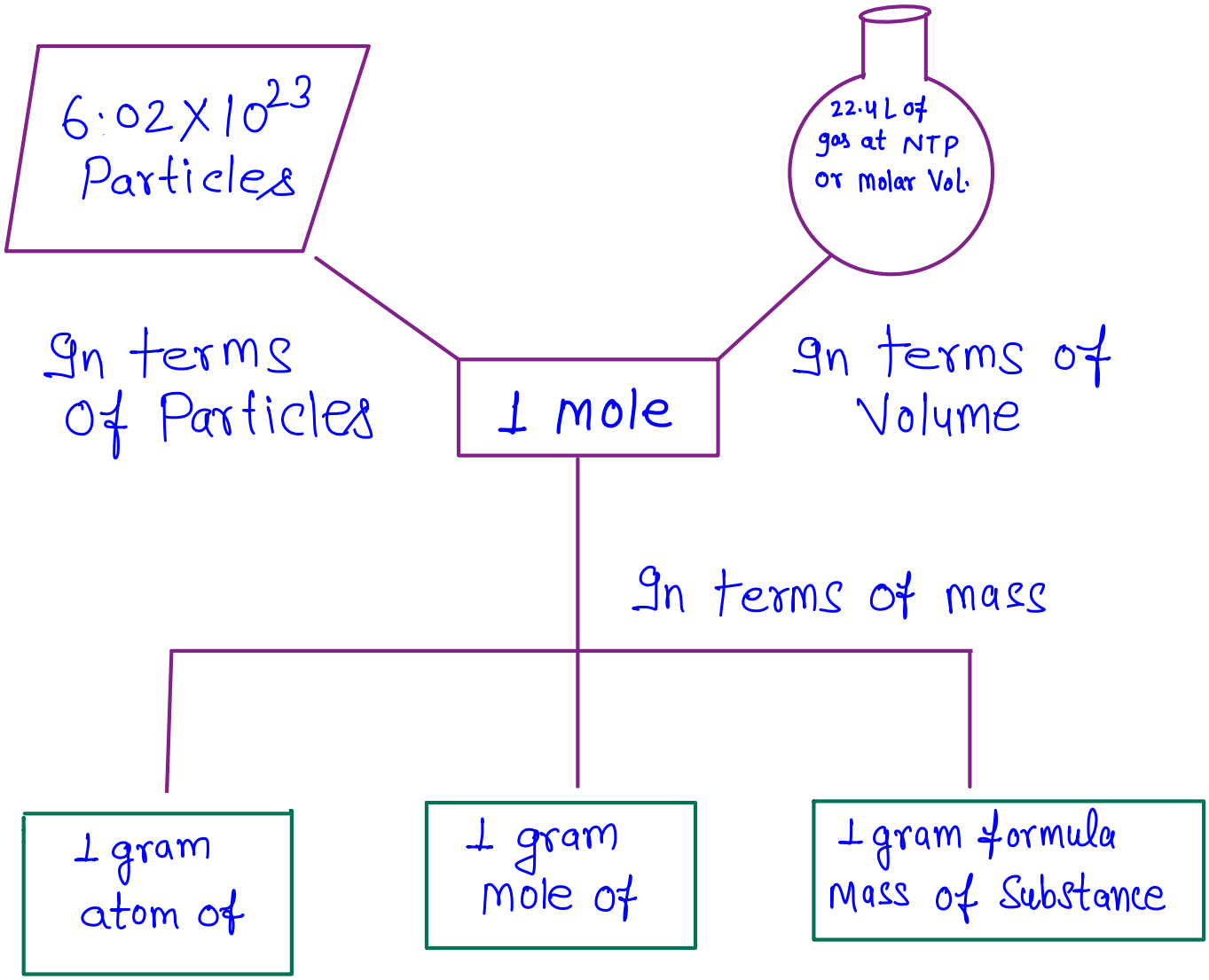
$$\Rightarrow \frac{\text{Mass of one atom of the element}}{\text{Mass of } \frac{1}{12}^{th} \text{ Part of mass of C-12 atom}}$$

* Average Atomic mass :-

Elements are found in different isotopic forms (atoms of same element having different atomic mass).
So the atomic mass of any element is the average of all the isotopic mass within a given Sample.

$$\text{Average atomic mass} = \frac{\sum(\% \text{ abundance}) \times \text{atomic mass}}{100}$$

Mole Concept :- Mole is the number equal to Avogadro's number just like a doyen is equal to 12, a Century means 100.
($N_A = 6.022 \times 10^{23}$)



$$\text{No. of mole of Molecule} = \frac{\text{Wgt in gm}}{\text{Molecular mass}}$$

$$\text{No. of Moles of atoms} = \frac{\text{Wgt in gm}}{\text{Atomic mass}}$$

$$\text{No. of moles of Gases} = \frac{\text{Vol. at STP}}{\text{Standard Molar Volume}}$$

(Standard Molar Volume at STP = 22.4 Lit)

Example: The Largest number of Molecules is in:-

(a) 28 g of CO (b) 46 g of C_2H_5OH

(c) 36 g of H_2O (d) 54 g of N_2O_5

Sol:- 36 g H_2O = 2 Mole of H_2O while other option has Less Mole.

* Gram Atomic Mass OR Gram Atom:-

Atomic weight of an element in grams is Called as gram atomic mass of an element Gram atomic mass is the weight of 1 mole atom (6.023×10^{23} atoms) of the element it is also Called as 1 gram atom.

Aw of C = 12, GAW of C = 12g = 6.023×10^{23} atom of Carbon = 1 Mole atom of C = 1 gram atom of Carbon.

Percentage Composition of a Compound:-

Percentage Composition of a Compound is the relative mass of each of the Consitituent element in 100 Part's of it by mass. Molecular mass of Compound, as discussed earlier (in reaction 1.10) Can be Calculated as the Sum of the atomic masses of all the atoms which Constitute the Molecule of that Compound.

Example:- Molecular mass of Calcium Nitrate:-

$Ca(NO_3)_2$ = atomic mass of Ca + 2 x Atomic Mass of N + 6 x Atomic mass of O = 40 + 2x14 + 6x16

= 40 + 28 + 96 = 164 u

% of Calcium (Ca) = $\frac{\text{Mass of Ca}}{\text{Molecular mass of } Ca(NO_3)_2} \times 100 = \frac{40}{164} \times 100 =$ 24.39

% of Nitrogen (N) = $\frac{\text{mass of N}}{\text{molecular mass of } Ca(NO_3)_2} \times 100 = \frac{28}{164} \times 100 = 17.07$

% of Oxygen (O) = $\frac{\text{mass of O}}{\text{molecular mass of } Ca(NO_3)_2} \times 100 = \frac{96}{164} \times 100 = 58.54$

Example: How much of Copper is obtained from 100 g of Copper Sulphate ($CuSO_4$) ?

Sol:- Molecular mass of $CuSO_4$ = Atomic mass of Cu + atomic mass of S + 4 x atomic mass of O

= 63.5 + 32 + 4x16

= 159.5 u

Gram molecular mass of $CuSO_4$ = 159.5 g

Now, 159.5 g of $CuSO_4$ here Cu = 63.5 g

$\therefore$ 100 g of $CuSO_4$ have Cu = $63.5 \times \frac{100}{159.5} = 39.81$ g

Empirical Formula:-

It is the formula which expresses the smallest whole number ratio of the constituent atom within the molecule.

Ex:- A Compound Contains C = 71.23%, H = 12.95% and O = 15.81%. What is the empirical formula of the Compound.

Sol:-

Element's Name with their Symbol	Atomic mass	Weight in Compound	Relative No. of Atom	Simplest atomic Ratio	Empirical Formula
Carbon (C)	12	71.93	$\frac{71.93}{12} = 5.936$	$\frac{5.936}{0.988} = 6$	
Hydrogen (H)	1	12.95	$\frac{12.95}{1} = 12.95$	$\frac{12.95}{0.988} = 13$	$C_6H_{13}O$
Oxygen (O)	16	15.81	$\frac{15.81}{16} = 0.988$	$\frac{0.988}{0.988} = 1$	

Chemical Equation and Stoichiometry:

Symbol:-

Liquid ⟶ (l)

Gas ⟶ (g)

Solution ⟶ (sol)

Aqueous Solution ⟶ (aq.)

Precipitate ⟶ (↓)

Light (gases) ⟶ (↑)

Heating ⟶ $\triangle$

Hit and Trial Method :-

Eg. Write the chemical equation for the following reaction and balance the same by hit and trial method:-

Sol:.

The skeleton equation for the reaction is:-

$$Ca_3N_2 + H_2O \longrightarrow Ca(OH)_2 + NH_3$$

The balancing is done in the following steps:-

(1) Balancing the Ca atoms

$$Ca_3N_2 + H_2O \longrightarrow 3\,Ca(OH)_2 + NH_3$$

(2) Balance the N atom:-

$$Ca_3N_2 + H_2O \longrightarrow 3\,Ca(OH)_2 + 2\,NH_3$$

(3)- Balance the O atom:-

$Ca_3N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2NH_3$

upon checking, the H atoms are alredy balanced

∴ final balanced chemical equation is:.

$Ca_3N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2NH_3$

Balance the following by yourself.

(1) $Ca + H_2O \longrightarrow Ca(OH)_2 + H_2$

(2) $Al_4C_3 + H_2O \longrightarrow Al(OH)_3 + CH_4$

* Problem Involving mass-mass Relationship:-

Example: What mass of Copper oxide will be obtained by heating 12.35 g of Copper Carbonate ? (Atomic mass of Cu = 63.5)

Sol:- The chemical eqn for the reaction is:.

$CuCO_3 \xrightarrow{Heat} CuO + CO_2$

$63.5 + 12 + 48 \qquad 63.5 + 16$

$\Rightarrow 123.5\,g \qquad \Rightarrow 79.5\,g$

123.5 g of Copper Carbonate upon heating gives Cu = 79.5 g

1.0 g of Copper Carbonate upon heating gives Cu = $\frac{79.5}{123.5}$ g

∴ 12.35 g of Copper Carbonate upon heating give Cu =

$$= \frac{79.5}{123.5} \times 12.35 = 7.95\,g$$

Problem Involving Mass - Volume Relationship:-

- Example:- $KClO_3$ on heating decompose to give KCl and O_2. what is the Volume of O_2 at NTP liberated by 0.1 mole of $KClO_3$?

Soln:- The chemical eqn for the decomposition of $KClO_3$ is

$$2\,KClO_3 \xrightarrow{\text{Heat}} 2\,KCl + 3O_2$$

2 mol — 3 mol

$(3 \times 22.4L = 67.2L)$

2 mole of $KClO_3$ evolve O_2 at NTP = 67.2 l

1 mole of $KClO_3$ evolve O_2 at NTP = $\frac{67.2}{2}$ l

0.1 mole of $KClO_3$ evolve O_2 at NTP = $\frac{67.2}{2} \times 0.1 l = 3.36L$

Problem Involving Volume - Volume Relationship:-

Example:

1 Litre of Oxygen at NTP is allowed to react with 3 Liter of Carbon monoxide at NTP. Calculate the Volume of each gas Found after the reaction.

Sol'. - The reaction takes place according to the equation

$$2CO + O_2 \longrightarrow 2CO_2$$

2L 1L 2L

1L of O_2 reacts with CO = 2L

∴ Volume of CO Left unreacted = 3-2 = 1L

Volume of CO_2 formed = 2L

Concept of Limiting Reagent :-

The reactant which is totally Consumed during the Cousse of reaction and when it is Consumed reaction stops.

The Concept of Limiting reactant is applicable to reaction other than monomolecular i.e. when more than one type reactant involved for. E.g. $2KClO_3 \longrightarrow 2KCl + 3O_2$ there is no Limiting reactant

So determine the limiting reagent amount of all reactant and mole ratio of reactants must be known. In the ratio of mole of reactant A with respect to reactant B is greater than the ratio of the moles of A to moles of B for a balanced chemical eqn than B is the Limiting reactant.

Eg. 3.0 g of H_2 react with 29.0 g O_2 to yield H_2O

(1) What is limiting reactant .. ?

(2) Calculate the maximum amount of H_2O that can be formed

(3) Calculate the amount of one of the reactant which remains unreacted.

Sol. The balanced chemical eqn is

$$2H_2 + O_2 \longrightarrow 2H_2O$$
$$2\times 2 = 4g \quad 32g \qquad 2\times 18 = 36g$$

Step-I – Determination of limiting reactant

4g of H_2 react with O_2 = 32g

$\therefore$ 3g of H_2 react with $O_2 = \frac{32}{4} \times 3 = 24g$

but the amount of oxygen actually available = 29g

As Oxygen is available in excess, therfore, hydrogen is the limiting reactant.

Step-II - Calculation of the amount of water that can be formed.

4g of H_2 From H_2O in the reaction = 36g

$\therefore$ 3g of H_2 from H_2O in the reaction $= \frac{36}{4} \times 3 = 27g$

Step (III) → Calculation of the amount of oxygen Left unreacted.

4g of H_2 required O_2 for reaction = 32g

∴ 3g of H_2 required for reaction = $\frac{32}{4} \times 3 = 24g$

but Oxygen which actually Present = 29g

∴ Amount of Oxygen Left unreacted = 29-24 = 5g

* Method of Expressing Concentration of Solution :-

Molarity (M) :- The Molarity of a Soln is the no. of moles of Solute Present in one litre (1 dm^3) of the Soln.

$$M = \frac{w \times 1000}{m_1 \times V \text{ in ml}}$$

$$M = \frac{\text{No. of moles of Solute}}{\text{Volume of Soln (in litre)}}$$

w = mass of Solute, m_1 = Molar mass of Solute

V = Volume of Solution.

* Molality (m) :-

The molality is the no. of moles of Solute Present in one kg of Solvent.

$$m = \frac{W_1 \times 1000}{m_1 \times W_2 \text{ (in gram)}} = \frac{\text{moles of Solute}}{\text{mass of Solvent (in kg)}}$$

Relation between Molarity and Molality:-

$$m = \frac{1000 \times M}{1000 \times d - M \times M_1}$$ (where d = density of Soln)

Part Per million Parts (PPm):-

It is defined as the mass of the Solute Present in one million (10^6) Parts by mass of the Soln. Thus for a Solute A,

$$PPm_A = \frac{\text{Mass of A}}{\text{Mass of Solution}} \times 10^6$$

Relationship between molarity (M) and Molality (m):-

Molarity M means n moles of Solute are Present in 1000 cc. of the Solution.

If density of the solution is d g/cc,
mass of solution = 1000 grams.
mass of solute = MM_2 g [M_2 is molar mass of solute)

Mass of Solvent = $1000d - MM_2$ g

$$\text{Molality } (m) = \frac{M}{1000d - MM_2} \times 1000$$

$$= \frac{1000M}{1000d - MM_2}$$

$$m = \frac{M}{d - MM_2/1000}$$

or $$m\left(d - \frac{MM_2}{1000}\right) = M \text{ or } M\left(1 + \frac{mM_2}{1000}\right) = md$$

or $$M = \frac{md}{1 + mM_2/1000}$$

* Relationship between Molality (m) and mole fraction (X_2):-

Molality (m) = moles of Solute / mass of Solvent (in kg)

Mole Fraction of Solute (x_1) = moles of Solute / (mole of Solute + Moles of Solvent)

- n_1 = Mole of Solute
- n_2 = Mole of Solvent
- M_2 = Molar mass of Solvent (g/mol)
- w_2 = mass of Solvent (g)
- m = Molality of Solute
- X_1 = Mole Fraction of Solute

From molality:

$$m = \frac{n_1}{w_2/1000} = \frac{n_1 \times 1000}{w_2} = w_2 = \frac{n_1 \times 1000}{m}$$

$$n_2 = \frac{w_2}{M_2} = \frac{n_1 \times 1000}{mM_2}$$

Now, Mole fraction of Solute:

$$X_1 = \frac{n_1}{n_1 + n_2} = \frac{1}{1 + \frac{n_2}{n_1}} = \frac{1}{1 + \frac{1000}{mM_2}}$$

Final Relationship:

$$X_1 = \frac{1}{1 + \frac{1000}{mM_2}} \quad \text{or inversely} \quad m = \frac{1000}{M_2} \cdot \frac{X_1}{1 - X_1}$$

Relationship between Molarity (M) and Mole fraction (X_2)

$$\text{Mole Fraction}, X_2 = \frac{M}{(1000 - d - MM_2) / M_1 + M_2}$$

$$= \frac{MM_1}{M(M_1 - M_2) + 1000d}$$

Thus $$X_2 = \frac{MM_1}{M(M_1 - M_2) + 1000d}$$

or $$M = \frac{1000d \times 2}{X_1M_1 + X_2M_2}$$

E.g:- Calculate the molarity of a Soln Containing 0.5 g of NaOH dissolved in 500 cm^3.

Sol:- Weight of NaOH dissolved = 0.5 g

Volume of the Solution = 500 cm^3

Calculation of Molarity:-

0.5 g of NaOH = $\frac{0.5}{40}$ mole of NaOH

(∵ Molar mass of NaOH = 40)

$\Rightarrow$ 0.0125 Mole

Thus, 500 cm^3 of Soln Contain NaOH = 0.0125 Mole

∴ 1000 cm^3 of the Soln Contain NaOH = $\frac{0.0125}{500} \times 1000$

= 0.025 Mole

Hence, Molarity of the Soln = 0.025 M

E.g: Calculate the molarity of a Soln Containg 9.8 gm of H_2SO_4 in 250 cm^3 of the Soln.

Soln :- Mass of H_2SO_4 dissolved = 9.8 g
Vol. of the Soln = 250 cm^3

Calculation of Molarity

Mol. mass of H_2SO_4 = 98

$\therefore$ No of moles of H_2SO_4 = $\frac{\text{mass in g}}{\text{molar mass}} = \frac{9.8}{98} = 0.1$

250 cm^3 of the Soln Contain H_2SO_4 = 0.1 Mole

$\therefore$ 1000 cm^3 of the Soln Contain $H_2SO_4 = \frac{0.1}{250} \times 1000$

$\Rightarrow$ 0.4 Mole

Hence, Molarity of Soln = 0.4 M